Youxing Kong

AI for Science in Complex Geotechnical and Geohazard Systems | Chinese Academy of Sciences, Beijing

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Research focuses on AI for science in complex geotechnical and geohazard systems: inferring system states from sparse, heterogeneous, multiscale observations; combining physical laws and domain knowledge to learn evolutionary mechanisms; and advancing failure prediction, uncertainty analysis, and reliable engineering decision-making.

Research pathway: Observation → Understanding → Prediction → Decision-making | Approach: Field engineering × Geotechnical mechanics and testing × AI for science

Education

PhD Student in Geological Engineering — Institute of Geology and Geophysics, Chinese Academy of Sciences Sep 2025–Present

- Research focuses on AI for science in complex geotechnical and geohazard systems, including spatiotemporal learning, physics- and knowledge-guided modelling, long-term deformation forecasting, and failure prediction.

Master of Engineering in Geological Engineering — Institute of Geology and Geophysics, Chinese Academy of Sciences 2021–2024

GPA: 3.66 | Thesis: Mechanisms of landslide initiation in fissured expansive soils

- Investigated fissure propagation and slope failure in expansive soils through fracture mechanics tests, digital image correlation (DIC), electrical resistivity monitoring, microstructural characterisation, slope model tests, and field-based landslide mechanism analysis.

Bachelor of Engineering in Civil Engineering — China University of Geosciences (Beijing) 2017–2021

GPA: 3.68 | Class rank: 2/78 (top 3%)

Professional Experience

Engineer (Intermediate Professional Title), Beijing Urban Construction Exploration & Surveying Design Research Institute Co., Ltd. Jul 2024–Present

- Conduct safety monitoring, intelligent inspection, and research for urban infrastructure, including mobile scanning and surface defect inspection of metro tunnels, structural monitoring, and municipal utility surveys.
- Apply image recognition, generative image processing, and time-series forecasting to engineering monitoring data; contribute to research papers, technical reports, and AI-related research proposals.

Entrepreneurship, Kulianggong Studio Oct 2023–Present

- Co-founded and operate an AI-driven research and knowledge studio, exploring generative AI applications in research support, data analysis, research communication, and knowledge management.
- Design end-to-end processes from task analysis and AI workflow development to deliverable production, exploring lean, AI-enhanced knowledge services and professional workflows.

Research Projects

Key Participant, National Key R&D Program of China: New Materials and Technologies for the Prevention and Control of Expansive-Soil Landslides and Engineered Slopes (2019YFC1509903) Oct 2020–Jul 2024

- Investigate the hydro-mechanical behaviour of expansive soils, slope failure mechanisms, and engineering mitigation techniques, with emphasis on the effects of moisture variation and soil strength degradation on slope stability.

Key Participant, Second Tibetan Plateau Scientific Expedition and Research Program: Geohazards and Risks Associated with Major Engineering Disturbances in the South Asia Corridor–Himalayas–River and Lake Headwater Regions (2019QZKK0904) Sep 2022–Oct 2024

- Investigate the occurrence, spatial variability, and risk formation mechanisms of geohazards along major engineering corridors on the Tibetan Plateau; explore geohazard investigation and risk assessment methods for complex mountainous terrain.

Key Participant, National Natural Science Foundation of China: Microscale Controls on Shallow Landslides in Multiscale Fissured Expansive Soils (42572368) Jan 2026–Present

- Investigate microstructural evolution, fissure development, and multiscale hydro-mechanical coupling in expansive soils under wetting–drying cycles to elucidate the microscale controls on shallow landslides.

Participant, National Natural Science Foundation of China: Investigation of the In-Service Structural Performance of Metro Shield Tunnels Using Large-Scale Data (52378385) Jan 2024–Present

- Investigate the evolution of in-service structural conditions, surface defect characteristics, and recognition reliability in shield tunnels, focusing on structural safety assessment under complex environmental conditions.

Participant, National Key R&D Program of China: Integrated Slope Protection and Ecological Restoration in Areas Disturbed by Major Hydropower Projects in the Upper Reaches of Southwestern China's Rivers (2024YFF1307800) Jul 2025–Present

- Investigate slope deformation and failure mechanisms, geohazard risk evolution, and the interaction between engineering protection and ecological restoration under disturbances caused by major hydropower projects.

Engineering Practice

Shandong Institute of Geophysical and Geochemical Exploration — Atmospheric Deposition Monitoring and Analysis Jul–Sep 2020

- Participated in field monitoring, sampling, and analysis of atmospheric deposition, gaining experience in environmental geochemical surveys, sample quality control, and field data acquisition.

Chuxiong, Yunnan — MICP-Based Ecological Slope Stabilisation Demonstration Sep–Dec 2021

- Participated in a demonstration of phosphogypsum utilisation and ecological slope stabilisation using microbially induced calcium carbonate precipitation (MICP), including microbial cultivation, material mixing, mechanised spraying, planting, and site construction.

Yelu Expressway, Henan — Coal Gangue as Subgrade Fill Mar–Jun 2023

- Conducted compaction, vibratory compaction, and California bearing ratio (CBR) tests on coal gangue with different particle sizes and gradations; assessed its suitability as subgrade fill using a trial road section.

Zhangjiawan Depot — Unloading Monitoring of a Large-Span Steel Dome Aug 2024

- Participated in strain sensor installation on critical members, continuous stress and temperature monitoring, and data analysis to support safety control during staged unloading of a large-span steel dome.

Beijing Metro — Fire-Door Inspection and Surveying of Tunnel Cross-Passages Apr 2025

- Participated in fire-door inspection, engineering surveying, and condition data collection for tunnel cross-passages during maintenance windows and night shifts, gaining experience in field coordination and inspection in complex underground environments.

Selected Publications and Manuscripts

Publications

Kong, Y., Li, Z., Sun, K., Qi, Z., & Hu, R. (2024). Study on the extension mechanism of fissures in expansive soil based on fracture mechanics test method. *Engineering Geology*, *343*, 107778. DOI

Kong, Y., Li, Z., Qi, Z., Sun, K., & Hu, R. (2024). Study on sliding surface characteristics of expansive soil based on quantitative evaluation of microstructure. *Research Square* [Preprint]. DOI

Wang, S., Li, Z., Li, L., **Kong, Y.**, Su, W., Qi, Z., Hu, R., & Qi, S. (2026). A nonlinear viscoelastic-plastic creep damage model of frozen soil-rock mixture under different temperatures and block conditions: Unique sights from the resistivity perspective. *Engineering Geology*, *370*, 108831. DOI

Sun, K., Li, Z., Wang, H., **Kong, Y.**, Wang, S., Zhou, Y., & Hu, R. (2025). Integrated machine learning using autoencoder and multilayer perceptron approaches to predict stress-strain curves of frozen soil-rock mixtures. *International Journal of Geomechanics*, *25*(10), 04025229. DOI

Wang, Z., Wang, Y., Hou, Y., Wang, X., Chen, Y., & **Kong, Y.** (2025). 3D laser scanning combined with BIM model for construction quality inspection of large-span steel structures—A case study of the dome steel structure in Zhangjiawan Depot Comprehensive Utilization Project. In D. Bassir (Ed.), *Smart manufacturing*, *structural health monitoring and digital twins* (pp. 122—131). IOS Press. DOI

Qi, Z., Li, Z., Li, L., **Kong, Y.**, He, Z., & Zhou, Y. (2024). Study on the tensile characteristics of Brazilian split in remolded loess based on resistivity and DIC techniques. *Environmental Earth Sciences*, *83*, 377. DOI

Li, Z., **Kong, Y.**, Fu, L., Zhou, Y., Qian, Z., & Hu, R. (2021). Model test study on deformation characteristics of a fissured expansive soil slope subjected to loading and irrigation. *Applied Sciences*, *11*(22), 10891. DOI

Kong, Y., Tang, C., Wang, X., Wang, Y., & Chen, Y. (2025). Dynamic stress prediction during unloading of a steel dome using an LSTM–seasonal differencing enhanced model. *Bulletin of Surveying and Mapping*, (S1), 104–109. (in Chinese) DOI

Kong, Y., Tang, C., et al. (2025). Deep learning-based recognition of water leakage defects in tunnel cross-sectional scanning grayscale images. 10th Conference on Rail Transit Surveying and Investigation Technology, March 31, 2025, Beijing. (in Chinese)

Sun, K., Li, Z., **Kong, Y.**, Zhou, Y., & Wang, S. (2023). Study of dynamic elastic modulus and cumulative plastic deformation in frozen soil-rock mixtures under uniaxial cyclic loading. *Journal of Glaciology and Geocryology*, *45*(6), 1730–1743. (in Chinese) DOI

Completed Manuscripts

Kong, Y., Li, Z., Li, L., Wang, S., & Hu, R. (2026). *A Single Image Is All You Need: A Knowledge-Guided Multimodal AI Framework for Geological Hazard Investigation*. [Under review; journal: Advanced Engineering Informatics].

Kong, Y., & Li, Z. (2026). *Where are expansive soils distributed in China? Evidence from digital soil mapping*. [Under review; target journal: Engineering Geology].

Kong, Y., Tang, C., Wang, Y., Li, Z., & Xu, Y. (2026). *Image enhancement for tunnel lining surface defect recognition: A conditional diffusion-based image inpainting approach for structural occlusion removal*. [Under review; target journal: Engineering Applications of Artificial Intelligence].

Kong, Y., Li, Z., Li, L., Wang, S., Hu, F., & Hu, R. (2026). *From sparse, asynchronous GNSS observations to continuous three-dimensional displacement fields: Multi-horizon spatiotemporal prediction for open-pit mine slopes*. [Manuscript prepared; target journal: Landslides].

Kong, Y., Li, Z., He, Z., Li, L., Wang, S., Hu, F., & Hu, R. (2026). *Atomistic mechanisms underlying structural hysteresis in expansive soils during wetting–drying: Evidence from molecular dynamics simulations of montmorillonite*. [Manuscript prepared; target journal: Engineering Geology].

Kong, Y., Li, Z., Li, L., Wang, S., & Hu, R (2026). Ontology-guided KG-RAG for fissured expansive-soil landslides: From knowledge organization to engineering application. [Manuscript prepared; target journal: Engineering Geology].

Selected Awards

- Outstanding Paper Award, 10th Conference on Rail Transit Surveying and Investigation Technology (2025).
- National Scholarship; First-Class Academic Scholarship (seven awards).
- Second Prize, China Undergraduate Mathematical Contest in Modeling; Honorable Mention, Mathematical Contest in Modeling (MCM).

Methods and Tools

AI-Augmented Development	Codex, ChatGPT, and other AI development tools; AI-assisted programming, research workflow automation, data engineering, and scientific computing; Python, C++, MATLAB, GIS, and numerical simulation software.
AI for Science	AI research workflow design and automation; research agents and multimodal large models; knowledge augmentation and intelligent reasoning; computer vision and generative AI; spatiotemporal intelligence and geospatial machine learning; physics-informed machine learning; digital soil mapping.
Geotechnical Engineering	Engineering geological investigation, geohazard assessment, slope stability analysis, geotechnical laboratory testing, microstructural characterisation (XRD, SEM/EDS, MIP, BET), three-dimensional laser scanning, and engineering monitoring (GNSS, DIC).

Science Communication

- Authored 20 popular-science articles on engineering geology for official Chinese Academy of Sciences platforms, attracting more than 200,000 cumulative reads.

Languages: Chinese (native); English: IELTS 7.5 (Listening 8.5 / Reading 7.5 / Writing 7.0 / Speaking 6.5).